AP Stats Lesson 3.4 Homework

Answer Key

1. a. The probability that a randomly selected student in Brett's first hour has a sibling is $\frac{24}{27} \approx 0.889$.

   b. The probability that a randomly selected student in Brett's first hour has no pets is $\frac{12}{27} \approx 0.444$.

   c. The probability that a randomly selected student in Brett's first hour has a sibling or a pet is $\frac{26}{27} \approx 0.963$.

   d. No, 13 of the 27 students have both a sibling and a pet. Since both events can occur at the same time, they are not mutually exclusive.

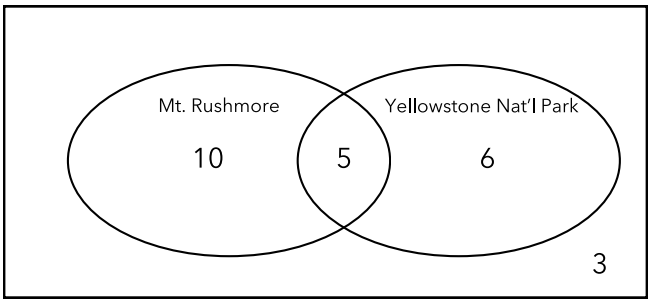
2. a. Yes, these events are mutually exclusive because they are complements and therefore cannot occur at the same time.

   b. No, these events are not mutually exclusive because someone could have a bachelor's degree and a doctorate degree at the same time.

   c. No, these events are not mutually exclusive because someone who has a bachelor's degree would have attended college so both events can happen at the same time.

   d. Yes, these events are mutually exclusive because a student can only be enrolled in one 9 AM class, so these events cannot occur at the same time.

3. a.



   b. $\frac{5}{24} = 0.208$

   c. $\frac{11}{24} = 0.458$

   d. $\frac{21}{24} = 0.875$

   e. $\frac{6}{24} = 0.25$

4. 2

5. a. $P(\text{6-9 years and AP math class}) = P(\text{6-9 years} \cap \text{AP math class}) = \frac{5}{66} \approx 0.076$

   b. $P(\text{AP math class}) = \frac{34}{66} \approx 0.515$

   c. $P(\text{0-2 years}^C) = 1 - \frac{16}{66} = \frac{50}{66} \approx 0.758$

   d. $P(\text{0-2 years or Non-AP math class}) = P(\text{0-2 years} \cup \text{Non-AP math class}) = \frac{16}{66} + \frac{32}{66} - \frac{13}{66} = \frac{35}{66} = 0.530$

6. D

7. C